Exact Dependence Geometry for Finite First Passage

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Abstract

We derive exact dependence geometry for all 400 ordered pairs of twenty frozen first-passage variables. Two-dimensional finite differences give 115600 joint PMF cells, exact covariance and Pearson-square data, total variation and L1 product discrepancies, and Frechet interval geometry. The relation spectrum is 20 diagonal, 164 strict-comparable, and 216 incomparable pairs. All claims are finite probability statements under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Joint law and dependence

Let $S_{ij}(k,\ell) = \#\{T_i > k, T_j > \ell\}$. The exact joint count is

$$N_{ij}(a,b) = S_{ij}(a-1,b-1) - S_{ij}(a,b-1) - S_{ij}(a-1,b) + S_{ij}(a,b).$$

From N_{ij} we compute $\mathrm{Cov}(T_i,T_j)$, ρ_{ij}^2, total variation from $P_i \otimes P_j$, the corresponding L1 distance, and the cellwise Frechet interval width and violation.

Theorem 1. *All 400 joint laws are nonnegative, normalized, and recover both C88 marginals. Transpose cells agree exactly; diagonal cells lie on the identity and have covariance equal to the target variance. Every cell obeys its Frechet interval, so the violation ledger is zero.*

Proof. Finite Mobius inversion gives the joint PMF and telescoping gives the marginals. Transposition follows from exchanging the two thresholds. The diagonal is the law of (T_i, T_i), and Frechet bounds are the elementary two-event bounds applied cellwise. □

certified object	count
ordered target pairs	400
joint PMF cells	115600
covariance/correlation records	400
TV and L1 product discrepancies	400 / 400
Frechet cell records	115600

2 Scope

No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside/table-of-marks, or Hilbert–Pólya operator is claimed.